Modeling the Effects of Electrode Shape and Separation Distance on Dendrite Growth in Solid-State Batteries

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U.S. DEPARTMENT OF ENERGY — Office of Science

Abstract

This project aims to develop a preliminary model to understand dendrite growth in solid-state electrochemical cells by examining how current density distribution is influenced by electrode shape, separation distance, and electrolyte composition. Regions of high current density are preferential for dendrite initiation. Using COMSOL Multiphysics, we model three electrode geometries and plot the current density in a horizontal slice. We find that pointed and square electrodes produce maximum current densities approximately 300 and 3 times higher than round electrodes respectively. The maximum current density in round and pointed electrodes decreases with increasing electrode separation distance, due to enhanced electric field spreading. These shapes also show a single, consistent current density maximum, making them ideal for studying dendrite initiation. In contrast, square electrodes' current densities increase with distance, likely due to amplified corner effects at larger separations, which increase local maxima and reduce edge uniformity. This suggests that thicker electrolytes may intensify such hotspots. Future work will use operando optical microscopy to validate these predictions against physical dendrite growth.

Background Knowledge

- **Solid-state batteries** (SSBs) are safer and more efficient than lithium ion batteries (LIBs)
- **Argyrodite-Li_6PS_5Cl** is a good solid electrolyte material (high ionic conductivity)
  - Enhanced with a **poly-TEGDMA** coating
- **Dendrites**: needle-like lithium structures grow from anode to cathode, cause battery failure
  - From **uneven lithium deposition** due to poor interfacial contact
  - Form in regions of **high current density**
- **Altering electrode shape alters current density distribution** for controlled dendrite growth

Objectives

How do dendrites form and grow in solid-state electrochemical cells?

Understand the effects of:

- **Solid electrolyte composition**,
- **Electrode shape**, and
- **Electrode separation distance**

On the distribution of and maximum current density in a solid-state electrochemical cell via mathematical modeling.

Future Work

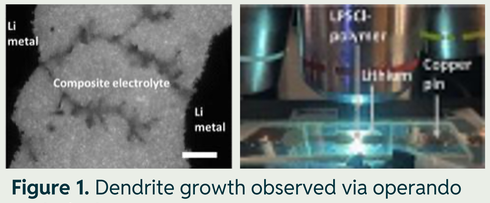


Figure 1. Dendrite growth observed via operando optical microscopy.

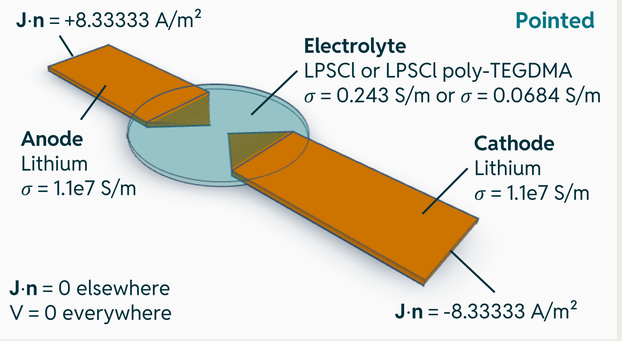
We will use **operando optical microscopy** to observe dendrite initiation sites in physical cells matching the modeled electrode geometries and observe if they align with high current density regions from our simulations.

Using the current density distribution data, we plan to **adapt existing dendrite growth models** to our 3D setup for more accurate predictions of dendrite propagation and the critical dendrite length for exponentiated growth.

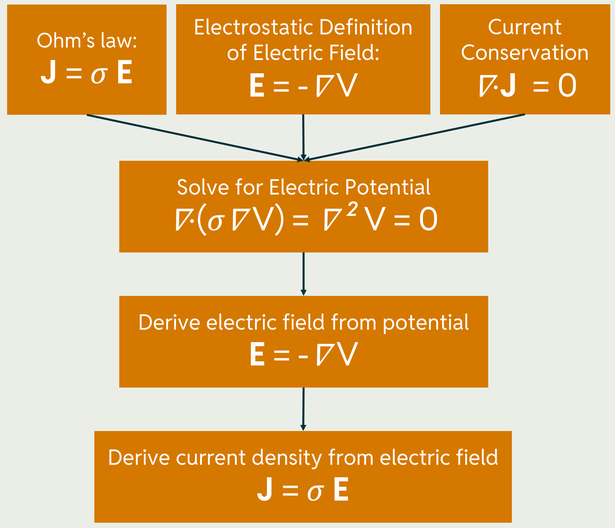
Finite Element Analysis

We use the COMSOL Multiphysics® Electric Currents Interface, which uses Maxwell's equations to simulate **steady-state current flow** in a solid-state electrochemical cell with **no volume current sources**. COMSOL solves partial differential equations via 3D finite element analysis by discretizing the geometry, solving for the electric potential, and computing current density.

Model Setup

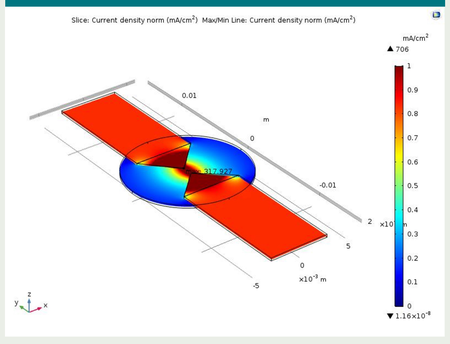


Computational Procedure

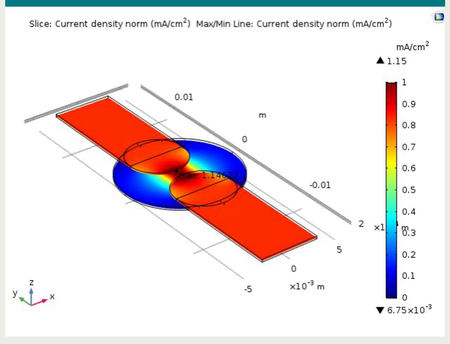


Plotting Current Density

Pointed



Round



Square

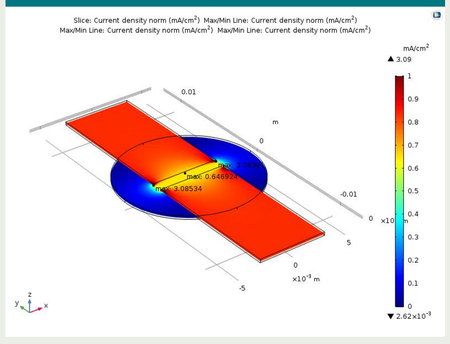
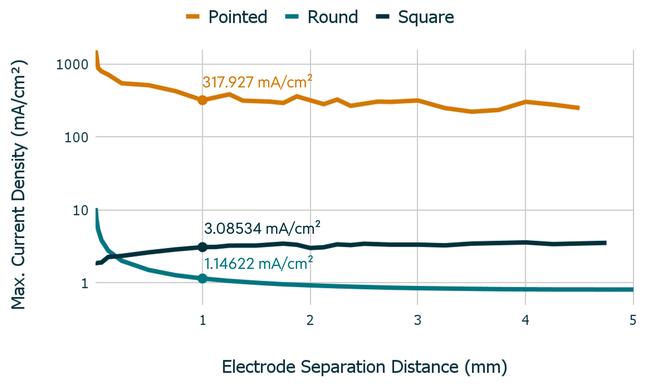


Figure 2. Colored surface plots of the current densities in cells with pointed, round, and square electrodes with 1 mm of separation and LPSCl solid electrolytes.

LPSCl



LPSCl poly-TEGDMA

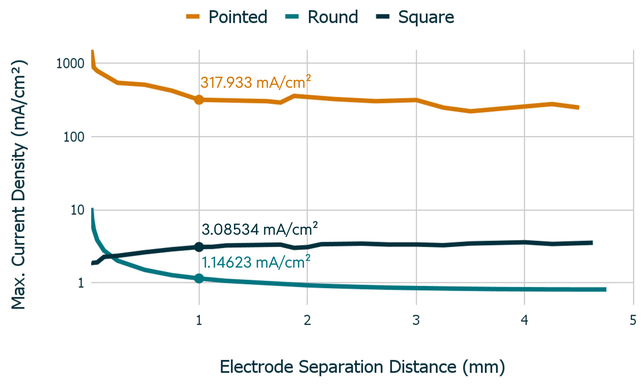


Figure 3. Maximum current density versus electrode separation distance for pointed, round, and square electrodes and electrolytes made of LPSCl and LPSCl poly-TEGDMA

Conclusions

Dendrite Prediction

- **Minimal difference between LPSCl and LPSCl poly-TEGDMA** current density distributions
- Dendrite growth promoted when cycled **beyond critical current density (~1 mA/cm²)**
- Beyond 1 mm separation, the **round electrodes maintain ~1 mA/cm²** max. current density
- For the same distances,
  - The **pointed** electrodes have max. current densities **300x higher**
  - The **square** electrodes have max. current densities **3x higher**

Model Selection

Circular and Pointed Electrodes

- **Increasing separation distance decreases peak current density**
- **Consistent** location of max. current density
- **Predictable** current density distribution, understand dendrite growth more easily

Square Electrodes

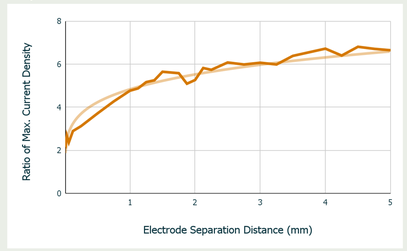


Figure 4. Ratio of current density at corner to center of electrode edge vs. separation distance.

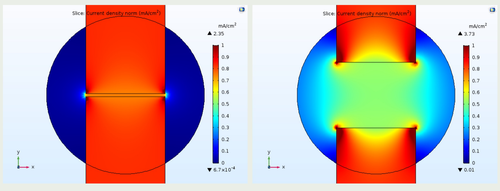


Figure 5. Current density for square electrodes with 0.25 mm and 5 mm separation and LPSCl electrolyte

- Less separation, current density is distributed uniformly along edge
- More separation, current density concentrated at corners

Manufacturing Considerations

- Importance of manufacturing methods like **thermal vapor deposition** for controlled surface morphologies without sharp dendrite-prone areas
- Further investigation of **optimal electrolyte thickness**, as thinner electrolytes may suppress high current densities at square electrode corners

Acknowledgements

This work was supported in part by the U.S. Department of Energy, Office of Science, Office of Workforce Development for Teachers and Scientists (WDTS) under the Science Undergraduate Laboratory Internship (SULI) program. Special thanks to Workforce Education and Development at Lawrence Berkeley National Lab and to my mentors Gao Liu, Alvaro Videla, and Xiuyu Jin for the valuable experience this internship has provided me with.

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